

Empowering the SOC with Security Copilot

Protect at the speed and scale of AI



Navigating the complex threat landscape



1,500+

threat actors tracked
by Microsoft



58%

of organizations use more
than 40 security tools



4M+

global shortage
of cybersecurity workers

We face an unprecedented threat landscape

Speed

Median time for an attacker to access private data from phishing (in 2022)



72 mins

Scale

Blocked password attacks per second

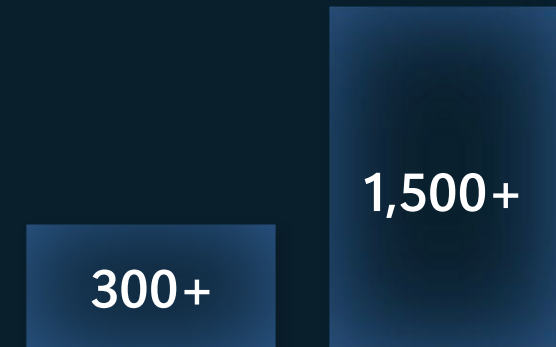


2023

2024

Sophistication

Threat actors tracked by Microsoft



2023

2024

*Extrapolated from 30B per month

Source: Microsoft Digital Defense Report 2022, Microsoft Digital Defense Report 2023 and Microsoft Digital Defense Report 2024

The SOC is the pressure point



Unmanageable

Annual growth in log data volumes is **250%**



Fragmented

Teams work across **40+ security tools** but still have gaps in their ability to do advanced analytics



Outpaced

Median time for an attacker to access private data from phishing is **1h 12mins**

Empowering security teams to defend at the speed and scale of AI

35%



increase in accuracy
by security novices¹

30%



reduction in MTTR with
three months of Copilot use¹

60%



efficiency gains on tedious tasks
like alert triage and reporting²

7%



increase in accuracy
by security professionals¹

22%



faster across SOC use cases
for security professionals²

97%



security professionals want
to use Security Copilot again²

¹ Generative AI and Security Operations Center Productivity, November 2024

² Randomized Controlled Trial for Security Copilot, Jan 2024

A shift to autonomous protection

From...

To...

Manual upkeep with the latest settings recommendations



Prevent



Continuous adjustment of settings

Response playbook design and manual intervention



Respond



Self-learning defense

Always responding to adversary moves



Protect



Proactive cut-off

Standalone prevention and protection strategies



Prevent



Active feedback loop

Harness AI

to accelerate SOC processes
and stay ahead of evolving
threats



Accelerate investigations
to remediate faster



Leverage autonomous agents



Generate actionable intelligence

Microsoft's approach to AI-powered cybersecurity

Making AI a force multiplier

Cyber skills and promptbooks

+

Evergreen threat intelligence

+

Plugins

+

Security-specific orchestrator

+

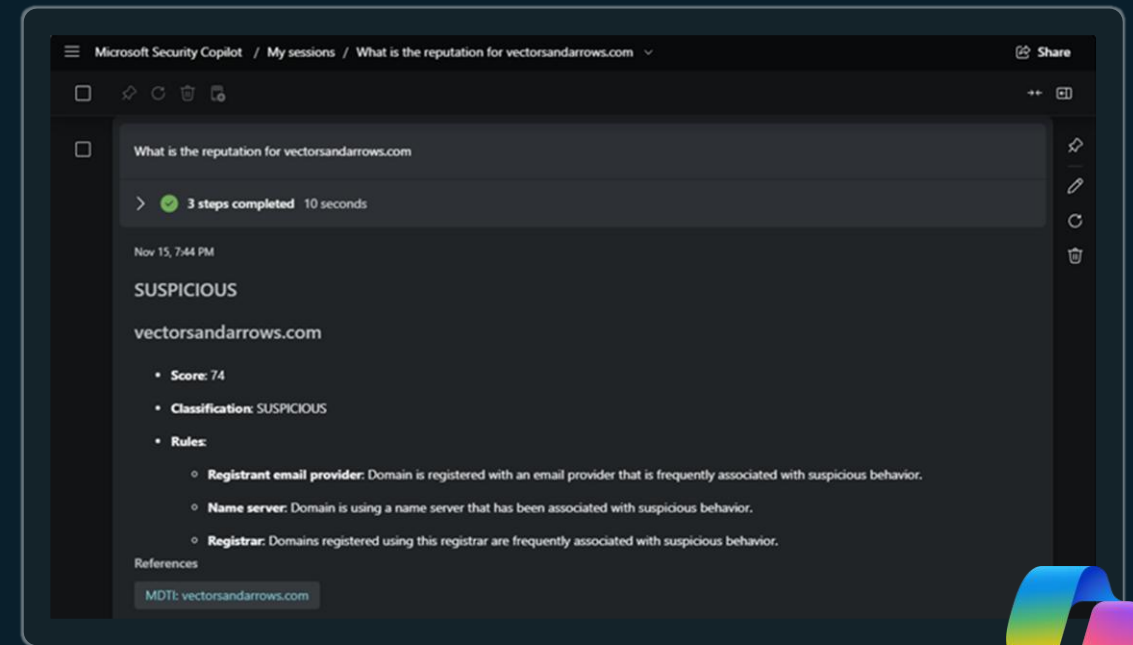
Hyperscale infrastructure

+

Advanced LLMs + Microsoft Security



Prompt: What is the reputation for vectorsandarrows.com?



with Security Copilot



Security Copilot skills

GenAI built for the SOC



Agentic AI

Security Copilot agents empower SOC analysts with always-on, autonomous defense, easing the burden of complex, high-volume security work

Incident summary



Copilot provides an overview of attacks, including timelines, assets involved, and indicators of compromise (IOCs)

Guided responses



AI-driven recommendations for incident response actions, categorized into triage, containment, investigation, and remediation

Script analysis



Analyzes scripts and code, helping security teams quickly identify if they are malicious or benign

Generating KQL queries



Converts natural language queries into KQL for advanced threat hunting

Creating incident reports



Instantly generates detailed reports summarizing incidents, actions taken, and a record of who performed each action and when

Threat intelligence



Delivers Microsoft Defender Threat Intelligence information about threat actors and tools

File analysis



Copilot provides a file's summary, including detection information, related file certificates, a list of API calls, and strings found in the file.

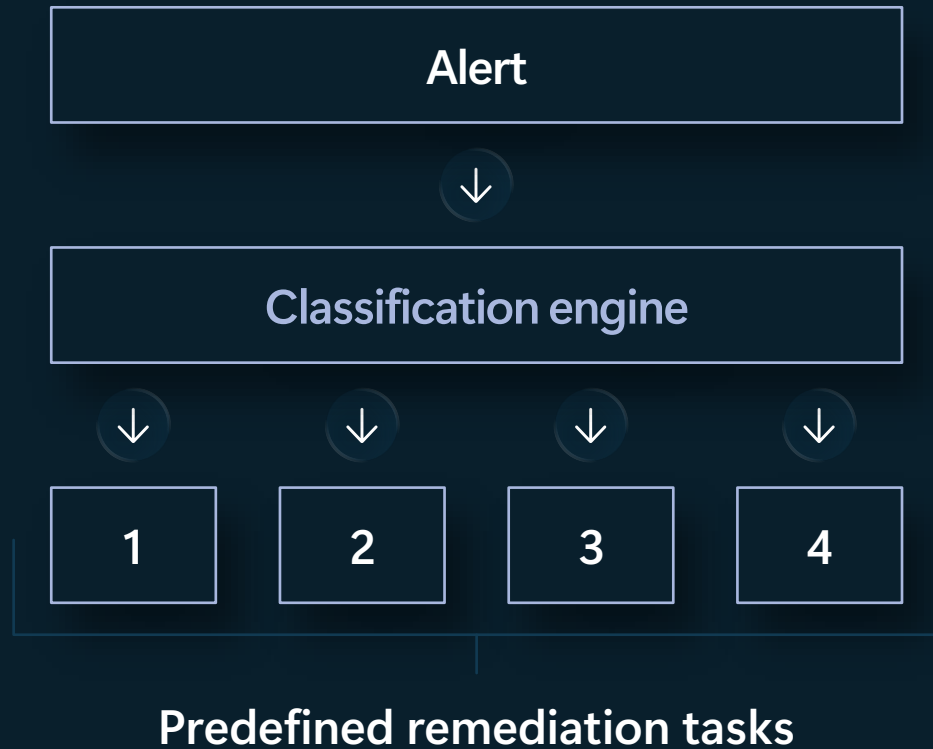
Identity and device summary



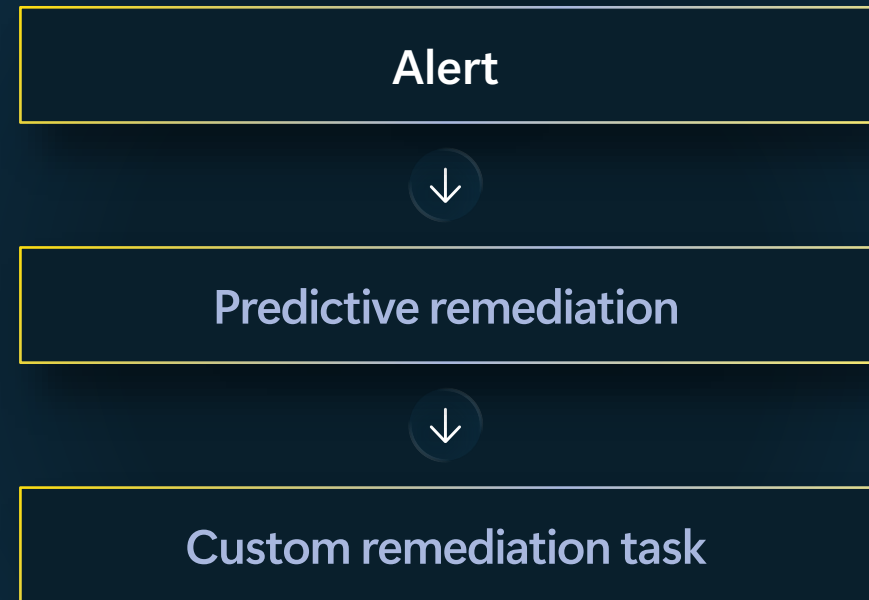
Provides summaries of suspicious identities with contextual insights, as well as device's security posture and any vulnerabilities or unusual behaviors

Security demands more than traditional automation

Rules-based approach



Generative AI

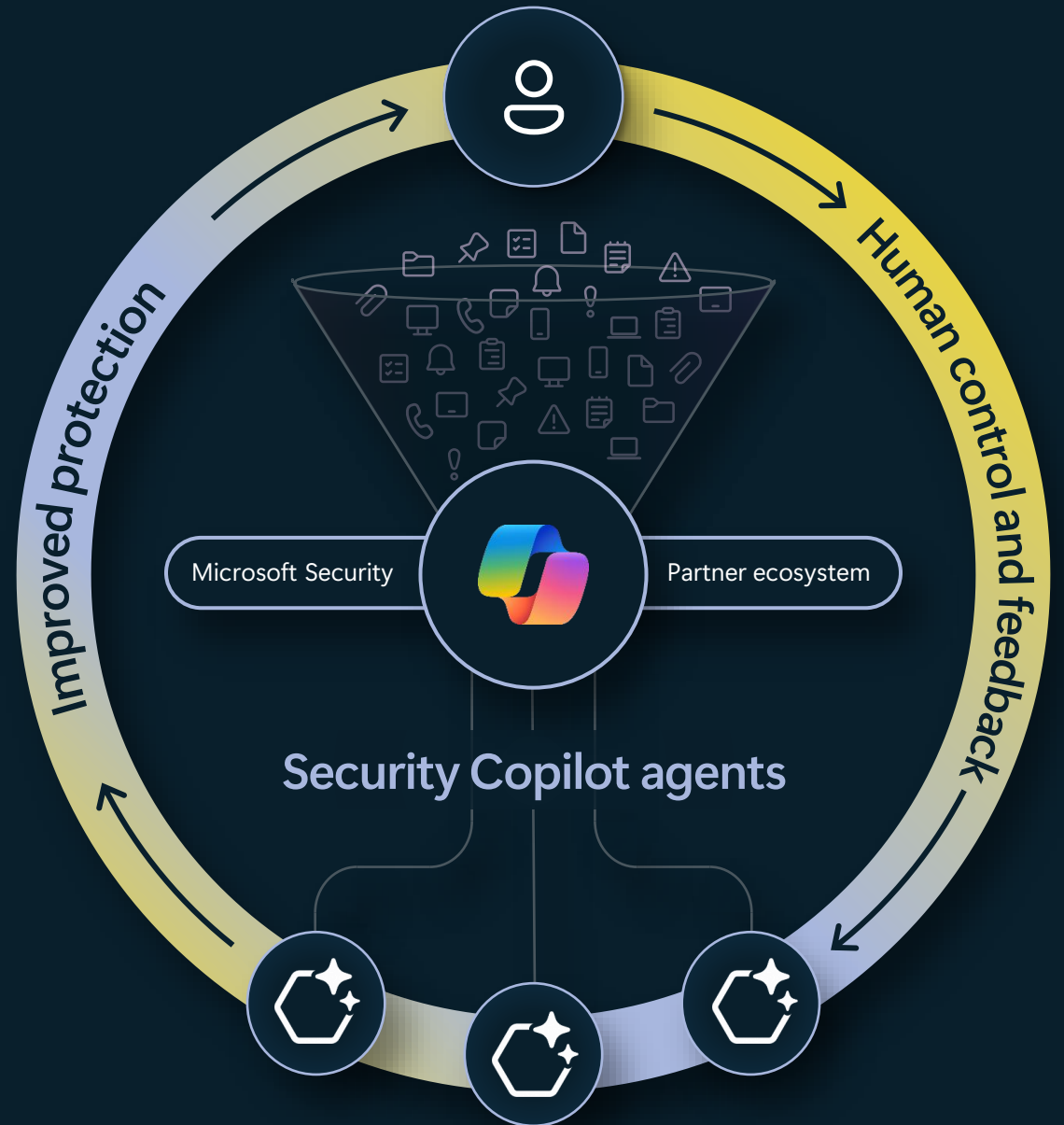


Security Copilot agents

Seamlessly integrating with Microsoft Security solutions and partner ecosystem, **Microsoft Security Copilot agents** enhance security and IT operations with autonomous and adaptive automation.

Purpose-built for security, agents:

- ✓ continuously learn from feedback
- ✓ dynamically adapt to organizational workflows with your team fully in-control
- ✓ operate securely within Microsoft's Zero-Trust framework



Threat Intelligence Briefing Agent

Threat insights mapped to you

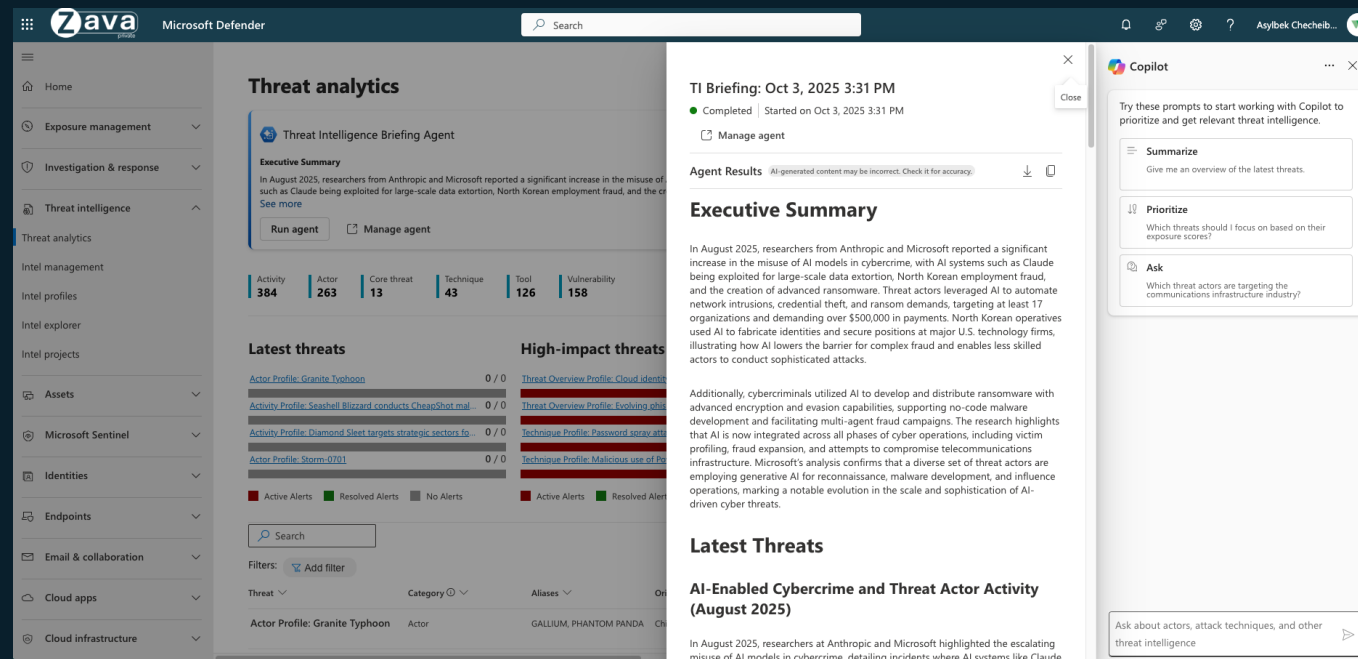
Provides the latest threat intelligence related to your organization's unique profile, including geolocation, IT infrastructure, and threat exposure.

Real-time situational awareness

Continuously learns from evolving attack landscapes and IT environments to provide the most relevant security updates.

Rapid reporting

Generates reports in minutes, eliminating the need for manual intelligence gathering and ongoing maintenance, which can take hours to days.

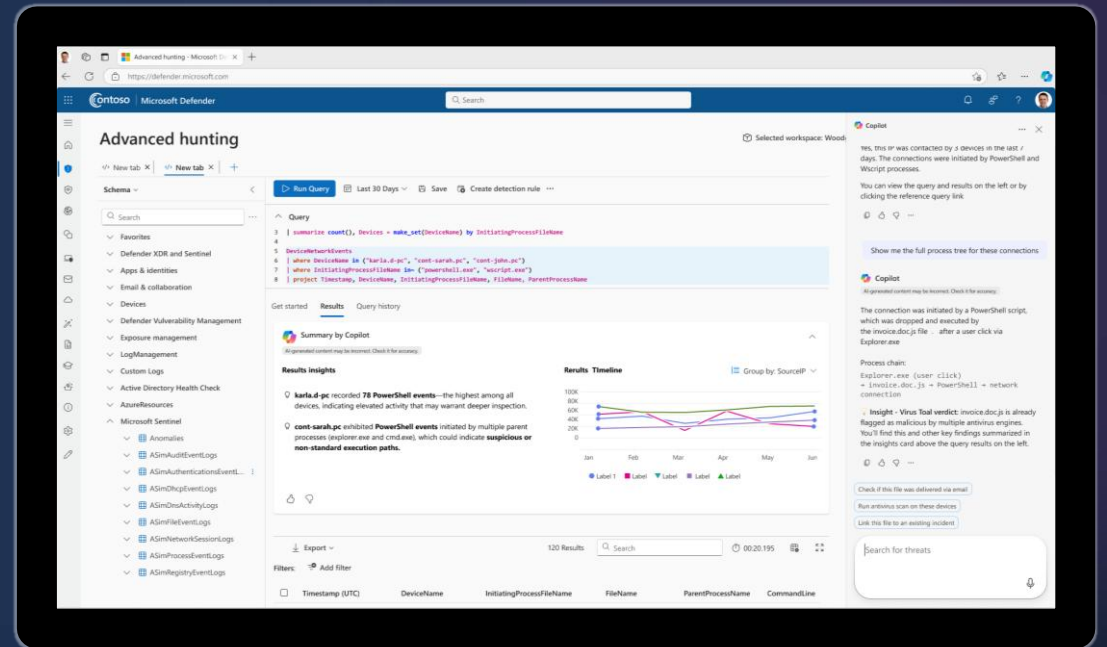


Vibe Hunting Agent

Only seasoned hunters could navigate complex query syntax — slowing investigations and leaving valuable insights buried in data.



Analysts investigate threats using natural language instead of complex query languages. An agent guides users through full hunting sessions with direct answers, insights, and suggested actions—making investigations faster and more accessible.



Dynamic Threat Detection Agent

Proactively uncover hidden threats

The agent continuously investigates incidents and telemetry to detect false negatives and blind spots, reducing missed threats and strengthening your security posture.

Continuously improve through feedback

When a gap is identified, the agent dynamically creates alerts enriched with risk details, MITRE mapping, and remediation recommendations—so analysts know exactly what to do next.

Builds a unified activity graph

Behind the scenes, it constructs a unified activity graph, connecting alerts, anomalies, and threat intelligence to surface behaviors that might otherwise go unnoticed

Copilot detection triggered **120 alerts** correlated to **48 incidents**.

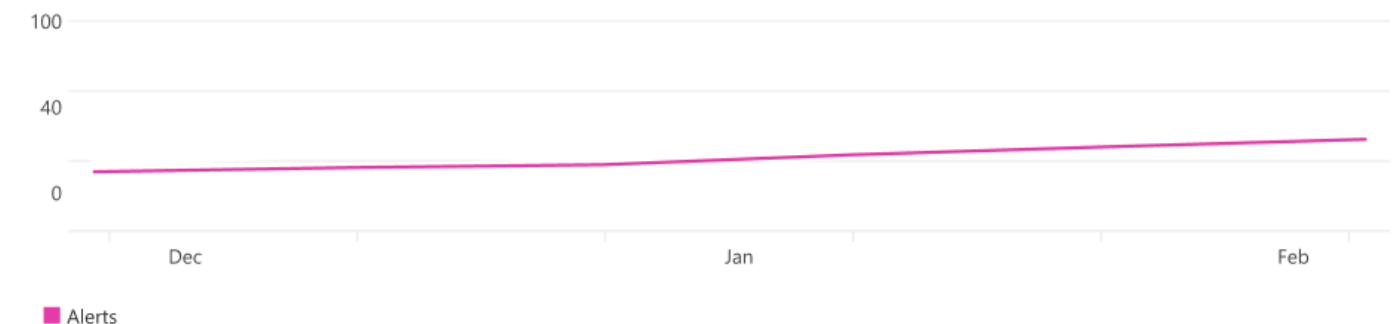
Copilot powered detections, leveraging the advanced AI capabilities embedded within the Copilot ecosystem.

AI detected alerts classified as true positive
44

Alert by severity



AI based Alerts over time



Copilot skills for the modern SOC

Copilot in the unified SOC platform



Intelligent context for incidents and alerts

Quickly assess emerging threats and your organization's exposure. Respond with enriched, AI-driven insights.

Rapid investigation and response

Copilot provides end-to-end support for analysts. From summaries of incidents and response, to assessment of incident impact, to actionable recommendations for faster investigation and remediation.

Unlock advanced SOC skills

Unlock new skills that allow analysts at all levels to complete complex tasks, like translating natural language to KQL or analyzing malicious scripts.

The screenshot displays the Microsoft Defender console interface. At the top, the title bar reads 'Microsoft Defender'. Below it, the breadcrumb navigation shows 'Incidents > Hands-on keyboard attack was launched from a compromised account (attack disruption)'. The main header area displays the incident title 'Hands-on keyboard attack was launched from a compromised account (at...)' along with a 'Copilot' button and a 'Manage incident' link. Below the header, there are tabs for 'Attack story', 'Alerts (44)', 'Assets (11)', 'Investigations (6)', 'Evidence and Response (70)', 'Recommended actions (21)', 'Summary', and 'Similar incidents (0)'. The 'Alerts' tab is selected, showing a list of alerts on the left. The central area features an 'Incident graph' with a network diagram showing connections between various entities like 'parkcity-jump', 'parkcity-win10r', and 'parkcity-win10v'. On the right, there are several panels: 'Logged on users (last 30 days)' showing 'Most logons' and 'Newest logon' for 'jonaw'; 'VM details' showing 'Category', 'Subtype', 'Domain', 'OS', 'SAM name', and 'Health state'; and a 'Copilot' panel with an 'Incident summary' and 'Guided response' section. The 'Incident summary' provides a detailed overview of the attack, including the timeline and the affected assets. The 'Guided response' section shows the status of the incident and provides recommendations for containment and remediation.

Incident summary and investigation

Gain context for incidents, triage faster with actionable summaries and guidance

Junior SOC
analyst



Before Copilot

SOC teams spent significant time manually gathering context for incidents, making it challenging to triage complex security alerts and respond effectively.



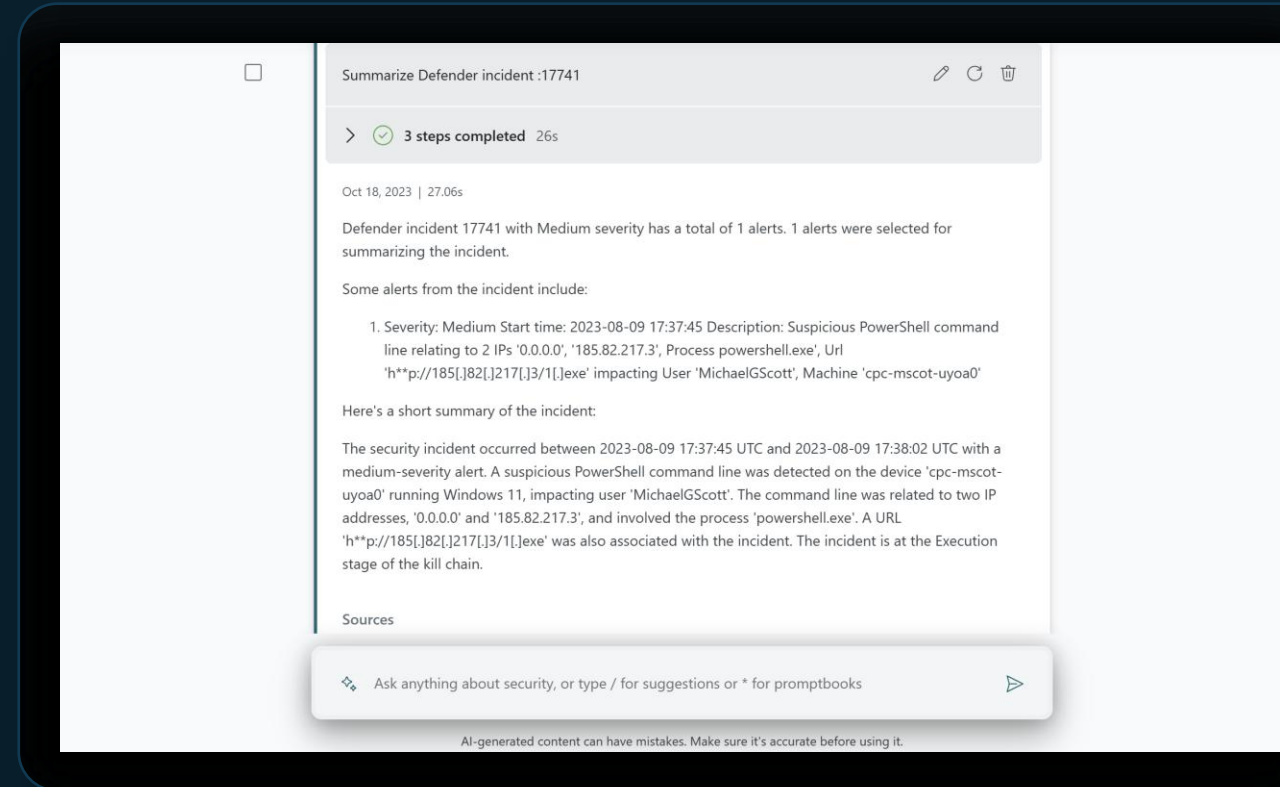
Prompt

Summarize Defender incident <xxxxx> in bullets.
Tell me about the entities associated with that incident.



Security Copilot

AI-generated, contextualized summaries so junior SOC analysts can make fully informed decisions.



Guided responses

Immediately triage and investigate incidents with recommended actions from Copilot

SOC
analyst



Before Copilot

Incident response was manual and inconsistent, with analysts relying on personal experience to determine appropriate actions, leading to potential delays and varied outcomes.



Prompt

Generate guided responses and recommendations for Defender incident {incident ID}



Security Copilot

Analysts can swiftly apply context-aware actions, ensuring consistent and efficient incident resolution, while also providing feedback to refine Copilot's future recommendations..

BEC financial fraud attack was launched from a compromised account (attack disruption)

Important: A potentially compromised account was disabled automatically by attack disruption in Microsoft 365 Defender. For more details, select the Alerts > Users tab or go to the [Action center](#).

Attack story | Alerts (20) | Assets (4) | Investigations (2) | Evidence and Response (25) | Recommended actions (21) | Summary | Similar incidents (0)

Alerts

- Mar 15, 2024 2:28 PM • Resolved
Activity from a Tor IP address
- Mar 15, 2024 2:28 PM • Resolved
Suspicious inbox manipulation rule
- Mar 15, 2024 2:28 PM • Resolved
Activity from a Tor IP address
- Mar 15, 2024 2:28 PM • Resolved
Suspicious inbox manipulation rule
- Mar 15, 2024 2:28 PM • Resolved
A potentially malicious URL click was detected
- Mar 15, 2024 2:28 PM • Resolved
A potentially malicious URL click was detected
- Mar 15, 2024 2:27 PM • Resolved
Anonymous IP address
- Mar 15, 2024 2:27 PM • Resolved
Anonymous IP address
- Mar 15, 2024 2:27 PM • Resolved
Compromised user account in a recognized attack pattern
- Mar 15, 2024 2:27 PM • Resolved
Anonymous IP address

Incident graph

2 Cloud Applications

Anonymous IP address

Update

Copilot

Completed recommendations 4/4

Status: All

Containment

- Completed
Disable the account Name
Attack Disruption
AI-generated content may be incorrect. Check it for accuracy.

Remediation

- New
Delete similar emails
We found emails that are very similar to emails involved in this incident. Delete these emails to contain the attack.
Soft delete emails | View similar emails
AI-generated content may be incorrect. Check it for accuracy.
- Completed
Disable the account Name in AD
AI-generated content may be incorrect. Check it for accuracy.
- New
Reset password for Name
The user account Name might be compromised. Require the user to change their password on their next sign-in.
Force password reset | View user
AI-generated content may be incorrect. Check it for accuracy.

Reverse engineering of scripts

Automate the reverse engineering of malware, enabling analysts to comprehend attacker actions

SOC and
TI analysts



Before Copilot

Junior SOC analysts struggle to understand complex scripts, which are time-consuming to analyze and hard to link to affected entities. Senior SOC analysts and TI analysts spend too much time manually analyzing malicious scripts, delaying action on identified threats.



Prompt

Analyze this script.
Extract the entities.
What threat intel is available for this URL?
Is it malicious, and if so, why?



Security Copilot

Security Copilot uses AI to automatically inspect and translate complex scripts into clear, actionable insights. This helps determine if scripts are malicious or benign and links indicators to affected entities, aiding in quicker and more accurate analysis.

Incidents > Human-operated ransomware attack was launched from a compromise... Copilot Manage incident

High In Progress Defender Experts
Feed Rate Ransomware Credential Phish Defender Experts Disruption Attack Disruption Device Not Onboarded HuntOR LATEST AlpineSkidhouse

Important! Attack disruption has automatically taken multiple response actions. For more details, go to the Action center.

Attack story Alerts (72) Assets (16) Investigations (9) Evidence and Response (127) Recommended actions (32) Summary Similar incidents (0)

Alerts
Play attack story Unpin all Show all
Apr 11, 2024 1:57 AM In progress
Suspicious URL clicked
percity-
win10x.parkcity.alpineshouse.co R
jones
Apr 11, 2024 1:57 AM In progress
Suspicious URL clicked on device
percity-
win10x.parkcity.alpineshouse.co
Apr 11, 2024 1:57 AM In progress
User accessed link in ZAP-
quarantined email
Jonathan Wolcott
Apr 11, 2024 1:57 AM In progress
PsExec launched a command on a
remote device
percity-
win10x.parkcity.alpineshouse.co R
jones
Apr 11, 2024 2:00 AM In progress
Compression of sensitive data
percity-
win10x.parkcity.alpineshouse.co R
jones
Apr 11, 2024 2:00 AM In progress
Possible attempt to access Primary
Refresh Token (PRT)
percity-
win10x.parkcity.alpineshouse.co R
jones
Apr 11, 2024 2:01 AM In progress
Active Directory Certificate Services
attack tool activity
percity-
win10x.parkcity.alpineshouse.co R

Incident graph Layout Group similar nodes
Communication Association
Back to incident details

PsExec launched a command on a remote device
Medium Detected In progress
Open alert page Manage alert

Details Recommendations
Remediation & prevention actions you should take
See recommended actions based on your alert

Script analysis
Apr 11, 2024 10:52 AM
The provided PowerShell script performs several actions including making a GET request to the URL "vectorsandarrows.com", finding and executing a file named "browsercore.exe", interacting with the Active Directory to manipulate certificate templates, creating and submitting a certificate signing request (CSR) using the "certreq.exe" utility, adding the current user to the local Administrators group, downloading a file from a specific URL, and extracting its contents, using the "rubocop.exe" utility to request a Kerberos ticket-granting ticket (TGT) and injecting it into the current session, establishing a network connection to a remote machine's C# share, and downloading and executing a file named "psexec.exe" from a specific URL, to remotely execute a PowerShell script on the remote machine.
1. The script starts by making a GET request to the URL "vectorsandarrows.com".
2. The script defines a function "get-userpntoken" that attempts to find and execute a file named "browsercore.exe" located in either the "ProgramFiles" or "win10x" environment variable paths. This function also sends a CSN request to a specific URL and reads the response.
3. The script calls the "get-userpntoken" function and writes the output to a file named "get11.json" in the temp directory.
4. The script defines a function "askforcert" that interacts with the Active Directory to find and manipulate certificate templates, and uses the "certreq.exe" utility to create and submit a certificate signing request (CSR).
5. The script calls the "askforcert" function with specific parameters, and then adds the current user to the local

Analyze
"powershell.exe" -EncodedCommand aQBuaHYAbwBrAGUALQByAGUAcwB0AG 0AZQB0AGgAbwBkACAALQBtAGUAdABo
Analyze
invoke-restmethod -method get -uri vectorsandarrows.com -usebasicparsing

KQL queries from natural-language

Leverage a query assistant that converts any natural-language question, in the context of threat hunting, into a ready-to-run KQL query.

Junior SOC
analyst



Before Copilot

Threat hunters and analysts needed to manually craft Kusto Query Language (KQL) queries, a process that was time-consuming and required specialized expertise



Prompt

In the Copilot prompt bar, users can enter natural language queries, such as: "Find all devices that signed into a risky session in the last week"



Security Copilot

Copilot translates the natural language input into a corresponding KQL query, allowing analysts to review, modify, and execute the query efficiently, thereby streamlining the threat hunting process

Advanced hunting

New query +

Schema Functions ...

Search

Alerts & behaviors

- AlertEvidence
- AlertInfo
- BehaviorEntities
- BehaviorInfo

Apps & identities

- AADSignInEventsBeta
- AADSpnSignInEventsBeta
- CloudAppEvents
- IdentityDirectoryEvents
- IdentityInfo
- IdentityLogonEvents
- IdentityQueryEvents

Email & collaboration

- EmailAttachmentInfo
- EmailEvents
- EmailPostDeliveryEvents
- EmailUrlInfo
- UrlClickEvents

Run query Last 24 hours Save Share link

Query

Query results are presented in your local time zone as per settings. Kusto filters, however, work in UTC.

1

Getting started Results Query history

Run basic queries Run advanced queries

Basic queries Advanced queries

Limit Shows 10 rows from the specified table Summarize

Where Apply filters to specific columns Extend

Copilot

Try these prompts to start hunting for threats with Copilot, or write your own natural language question to generate a KQL query.

- Display all service principal sign in logs where the location is "USA"
- What are all AADUserRiskEvents in the last 5 days where the risk level was medium?
- Show the top 5000 network connection events where the action type was "ConnectionRequest" and the protocol used was "Tcp" between 2022-10-01 and 2022-10-31.
- Here's a query you can use to find what you need:
`DeviceNetworkEvents | where Timestamp between (datetime('2023-04-03T13:59:43.0086815Z')) .. datetime('2023-05-17T16:42:00.0000000Z')`

Run query

Task automation

Get detailed analysis on any user in an incident using promptbooks with Logic Apps in Defender XDR and Microsoft Sentinel

Senior SOC
analyst



Before Copilot

For each security incident, SOC teams need to understand the users and impacted devices/applications. To automate this process in the past required a lot of code.



Action

In the Sentinel portal > Logic App designer, search for promptbook "Microsoft user analysis" to use Logic Apps to call on that promptbook within Copilot and automate adding user analysis details as comments to an incident.



Security Copilot

Copilot gathers data from various sources as instructed by each prompt in the promptbook, and adds all information to the incident activity log within Sentinel, saving time for security analysts.

The screenshot displays the Logic App Designer interface for a workflow named 'User-RiskCalc-Copilot'. The workflow consists of three main steps: 'Microsoft Sentinel entity', 'Parse JSON', and 'Run a Copilot for Security promptbook'. The 'Run a Copilot for Security promptbook' step is expanded, showing its configuration. The 'Promptbook Name' is set to 'Microsoft User Analysis'. The 'PromptbookInputs' section shows a JSON object with 'User_UPN' and 'duration' fields. The 'Advanced parameters' section shows 'Showing 0 of 1'. A detailed output window is visible, showing the results of the Copilot run, including user information, device details, login details, and audit logs.

Home > User-RiskCalc-Copilot

User-RiskCalc-Copilot | Logic app designer

Run a Copilot for Security promptbook

Parameters Settings Code view Testing About

Promptbook Name * Microsoft User Analysis

PromptbookInputs *

```
{  "User_UPN": "Natasha.David_0716@woodgrove.ms",  "duration": "7 days"}
```

Advanced parameters Showing 0 of 1

Comment created from playbook - User-RiskCalc-Copilot 10/23/24, 02:32 PM

Investigation Summary Report for Natasha.David_0716@woodgrove.ms

User Information

- User Principal Name (UPN): Natasha.David_0716@woodgrove.ms
- Display Name: Natasha David
- Department: Sales
- Role: Sales Rep I

Devices Associated

- Device 1: SUF_GO
- Device Name: SUF_GO
- Operating System: Windows 10
- Last Seen: 2024-10-22 22:45:33 UTC

Last Login Details

- Last Login Time: 2024-10-22 01:42:52 UTC
- Login IP Address: 2600:1700:6780:1900:d0a6:c4b2:9c61:387e
- Location: Mentor, Ohio, US

Authentication Methods

- Primary Method: Password
- Secondary Method: Microsoft Authenticator (iPhone 13)
- Additional Method: Windows Hello for Business (SUF_GO)

Audit Logs

- 2024-10-21 22:15:23 UTC: Add BitLocker key
- Result: Success
- IP Address: 2600:1700:6780:1900:d0a6:c4b2:9c61:387e
- Operation Type: Add
- User Agent: BitLocker/ 10.0.26100 (Windows BitLocker)



Thank you!